[**SUBMIT ANSWERS TO ALL QUESTIONS**]

Curves in parametric form. Tangent, normal and binormal vectors.

The length element of a curve $\boldsymbol{r}(t) = x(t)\boldsymbol{i} + y(t)\boldsymbol{j} + z(t)\boldsymbol{k}$ is $ds = |\boldsymbol{r}'(t)|dt$, where $\boldsymbol{r}'(t) \equiv d\boldsymbol{r}/dt$. The arc length between $t = a$ and $t = b$ is

$$s = \int_a^b |\boldsymbol{r}'(t)|dt = \int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2}\, dt.$$

Three unit vectors, $\boldsymbol{T}$ (tangent), $\boldsymbol{N}$ (normal) and $\boldsymbol{B}$ (binormal) for a curve in 3D are defined by

$$\boldsymbol{T} = \frac{d\boldsymbol{r}}{ds}, \qquad \frac{d\boldsymbol{T}}{ds} = \kappa \boldsymbol{N}, \qquad \boldsymbol{B} = \boldsymbol{T} \times \boldsymbol{N},$$

where κ is the *curvature* and $R = 1/\kappa$ is the *radius of curvature*. For a plane curve, $\boldsymbol{B} = \text{const}$, while in general, $\dfrac{d\boldsymbol{B}}{ds} = -\tau \boldsymbol{N}$, where τ is the *torsion*, and $\dfrac{d\boldsymbol{a}}{ds} \equiv \dfrac{1}{|\boldsymbol{r}'(t)|}\dfrac{d\boldsymbol{a}}{dt}$ for any vector $\boldsymbol{a}$.

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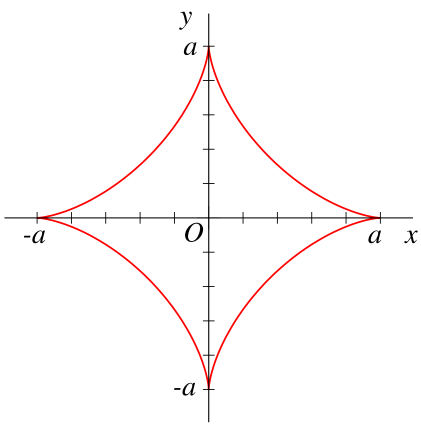
1. (a) Use integration by parts to show that [10]

$$\int \sqrt{1+t^2}\, dt = \frac{1}{2}t\sqrt{1+t^2} + \frac{1}{2}\ln\left(t + \sqrt{1+t^2}\right) + C.$$

[Hint: after integration by parts, aim to get $-\int \sqrt{1+t^2}\, dt$ on the right-hand side.]

(b) Find the arc length of the parabola $x = t$, $y = \frac{1}{2}t^2$ between points $t = 0$ and $t = 1$. [10]

2.



The curve shown is the *astroid*,

$$x^{2/3} + y^{2/3} = a^{2/3}.$$

Verify that it can be parameterised as [20]

$$x = a\cos^3 t, \quad y = a\sin^3 t, \quad 0 \le t \le 2\pi.$$

and find its arc length.

[Hint: the astroid consists of four identical segments.]

3. Find the arc lengths of the following space curves:

(a) $x = t$, $y = t^2$, $z = \frac{2}{3}t^3$, between points $t = 0$ and $t = 2$. [15]

(b) $x = e^t \cos t$, $y = e^t \sin t$, $z = e^t$, between points $t = 0$ and $t = t_0$. [15]

4. Find the tangent, normal and binormal vectors for the curve

$$x = e^t\cos t, \quad y = e^t \sin t, \quad z = e^t,$$

and determine its curvature κ, radius of curvature R and torsion τ. [30]